

Lab 3: Component Modelling & Architectural Pattern Selection

Objective

Evaluate different architectural styles, select the most appropriate one for the assigned scenario, and create a comprehensive UML Component Diagram showing modules, interfaces, and dependencies. This lab focuses on understanding how components interact within a system and how architectural decisions impact system design.

Duration

60 minutes

Software Requirements

- UML tool (draw.io, LucidChart, StarUML or any online UML tool)

Learning Outcomes for Lab 3

By the end of this lab, you will be able to:

1. Evaluate architectural styles:
 - Compare and contrast different architectural patterns (Layered, Microservices, Client-Server)
 - Analyse pros and cons of each pattern relative to specific scenarios
2. Select appropriate architecture:
 - Choose the most suitable architectural style based on system requirements
 - Justify architectural decisions with concrete reasoning
3. Design component diagrams:
 - Create UML component diagrams with proper notation
 - Show provided and required interfaces between components
 - Demonstrate understanding of component dependencies
4. Document architectural decisions:
 - Write clear, technical justifications for architectural choices
 - Explain security and performance implications

Introduction:

Component modelling Fundamentals

What is a Component?

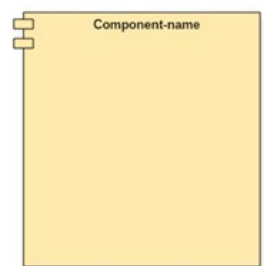
A component is a replaceable and executable piece of a system whose implementation details are hidden. Key characteristics:

- **Modular:** Encapsulates specific functionality
- **Replaceable:** Can be swapped with other components that provide the same interface
- **Black Box:** External behavior is defined by interfaces, internal implementation is hidden
- **Executable:** Participates in system execution

Component Diagram Elements

1. Component Notation

- Represented as a rectangle with <<component>> stereotype
- Component name placed at center
- Optional component icon in upper-right corner
- Implementation details are hidden from external view



2. Interfaces

Provided Interface (Ball notation):

- Services that a component offers to other components
- Shown as a circle connected to the component
- Represents what the component can do for others

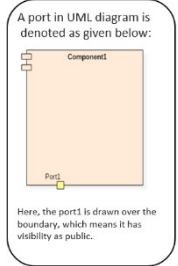


Required Interface (Socket notation):

- Services that a component needs from other components
- Shown as a semicircle connected to the component
- Represents what the component needs from others

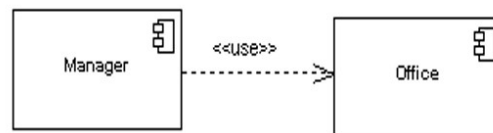
3. Ports

- Interaction points between components and external environment
- Group related provided and required interfaces
- Can be public (drawn on boundary) or private (drawn inside)

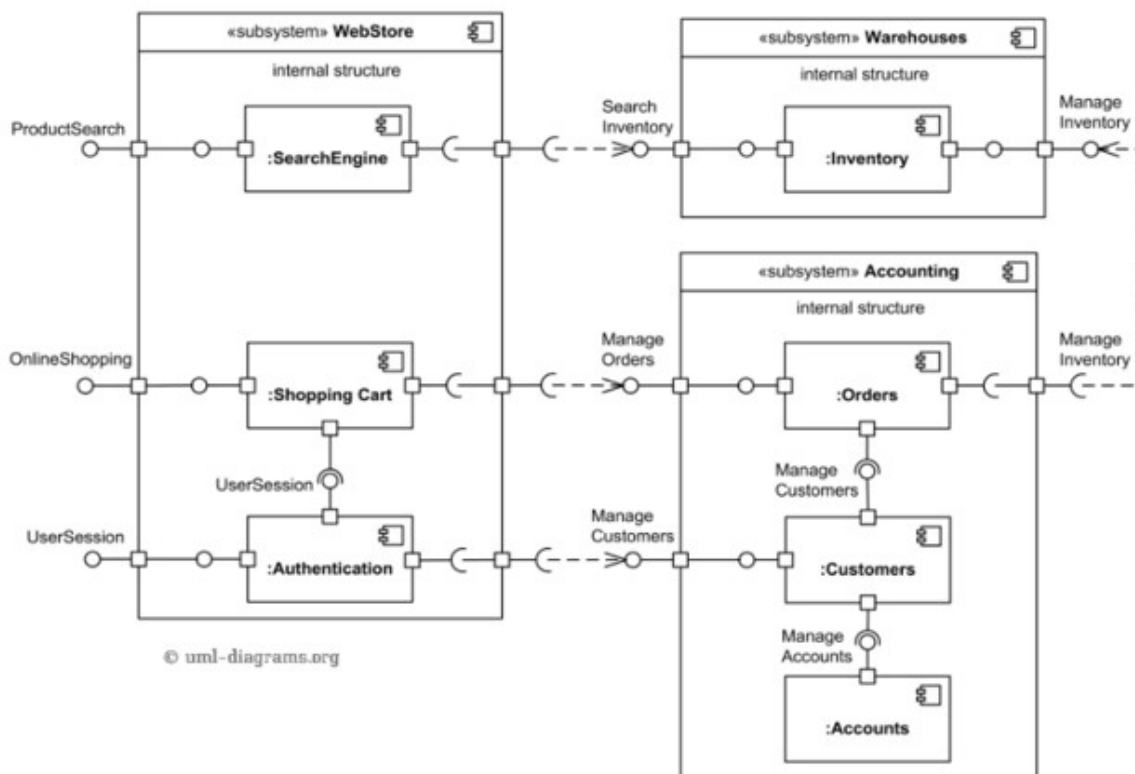


4. Dependencies

- Usage Dependencies: Shown as dashed arrows with <<use>> stereotype
- Assembly Connectors: Connect provided interfaces to required interfaces



Example Component diagram:



Online shopping UML component diagram example with three related subsystems - WebStore, Warehouses, and Accounting.

Architectural Styles Overview

1. Layered Architecture

- **Structure:** Organized in horizontal layers (Presentation, Business, Data)
- **Pros:** Clear separation of concerns, easy to understand, good for traditional applications
- **Cons:** Performance overhead, tight coupling between layers, difficult to scale individual components

2. Microservices Architecture

- **Structure:** Small, independently deployable services
- **Pros:** Independent scaling, fault isolation, technology diversity, independent deployment
- **Cons:** Operational complexity, network latency, data consistency challenges

3. Client-Server Architecture

- **Structure:** Centralized server with multiple clients
- **Pros:** Centralized control, simple deployment, easy data consistency
- **Cons:** Single point of failure, scalability bottleneck, limited fault tolerance

Example Scenario:

Self-Service Coffee Kiosk System

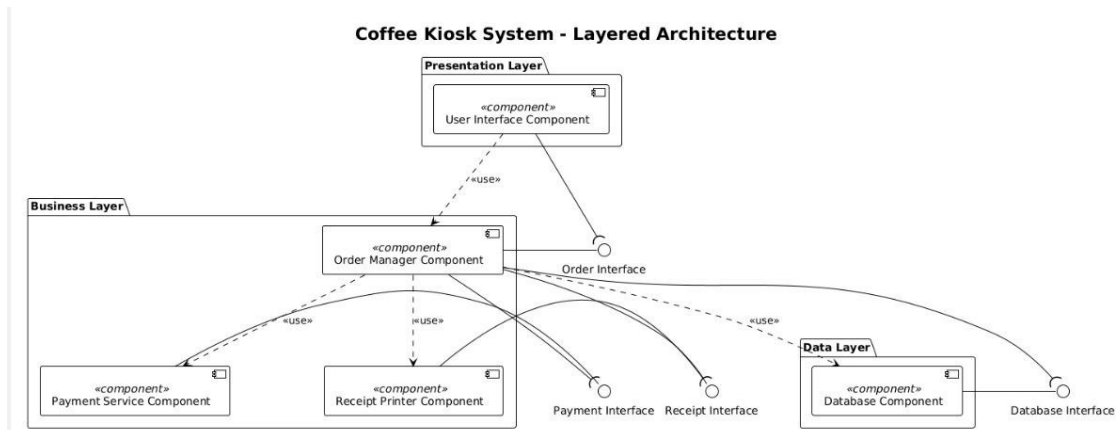
You are designing the software architecture for a Self-Service Coffee Kiosk system in a busy café. The kiosk must provide a complete coffee ordering experience with the following requirements:

Core Functionality:

- Customers can select from 3 coffee types (Espresso, Americano, Latte)
- Customers can choose from 2 drink sizes (Small, Large)
- Customers can pay via credit card only
- System prints receipts with order details

Technical Constraints:

- Must handle touch screen interface interactions
- Must connect to receipt printer hardware
- Must store menu data and pricing information



Deliverables

1. Component Diagram (PNG or PDF)

Your diagram must include at least 5 components (Adding attributes is optional).

Interface Requirements:

- Show at least 4 interfaces between components:
- Use proper UML notation for provided/required interfaces
- Show data flow and component interactions

2. Written Justification (Word document, max 1 page)

Lab 3 Steps

Step 1: Scenario Review

- Read the assigned scenario thoroughly
- Identify key functional and non-functional requirements
- List the main challenges (performance, usability, security)

Step 2: Architectural Style Analysis

Analysis: Analyse each architectural style.

Step 3: Architecture Selection & Component Identification

Select Architecture

Identify Components: Based on your chosen architecture, identify all necessary components (*Two components are given below, please identify the remaining 3*):

1. Order Manager Component
2. Payment Service Component

Step 4: Component Diagram Creation

Setup:

- Open your UML tool (draw.io recommended)
- Set up the canvas with adequate space

Draw Components:

- Create component rectangles for each of the 5 identified components
- Use <<component>> stereotype or clear component notation
- Arrange components logically on the canvas
- Group related components (e.g., customer-facing, data storage)

Add Interfaces: Define and draw the required interfaces between components:

Key Interfaces to Show:

Given:

- Order Manager ↔ Payment Service: Payment processing requests

Interface Notation Guidelines:

- Use solid lines for internal service communications
- Label interfaces with protocol/technology (API calls, hardware interfaces, database queries)
- Use provided interface (ball) and required interface (socket) notation where appropriate

Create a one-page Word document with the following structure:

Architecture Selection: "We chose [Architecture] for the <Assigned> System."

Your justification must include:

- Architectural Choice: State which architectural style you selected
- Two Reasons: Provide two specific, scenario-related reasons for your choice
- Security Advantage: Explain one way your architecture addresses security concerns
- Performance Benefit: Describe one performance advantage of your chosen architecture

Step 6: Diagram Finalization & Export

- Review diagram for completeness
- Ensure all required components are present
- Verify interface directions and labels are correct
- Check UML notation compliance
- Export as PNG or PDF
- Save justification document as PDF
- Submit the same through the Github repo (Ex: Lab3 under SELABS)